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BE (5th Semester)

Examination, April-May, 2013

Branch : Mechanical

INTERNAL COMBUSTION ENGINES

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Attempt all questions. Part (a) of all questions are compulsory. Answer any two parts from part (b), (c) & (d).

(ii) Explain your answer by means of neat sketches/graphs wherever it is necessary. Answer point-to-point.

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Q. 1. (a) Define "mean effective pressure" and "specific fuel consumption". 2

(b) Draw the valve timing diagram for a 4 stroke S.I. Engine. Explain why valves are not opened and closed at dead centres. Discuss the effect of VT on volumetric efficiency of engine. 7

(c) Discuss briefly the effect of compression ratio and mixture strength on thermal efficiency. Also discuss the effect of variation of specific heat on cycle efficiency. 7

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(d) What is Scavenging ? Explain exhaust blowdown loss. Discuss the optimum opening position of exhaust valve to reduce this loss. 7

Q. 2. (a) What is stoichiometric air-fuel ratio ? 2

(b) Discuss the effect of volatility on S.I. Engine performance briefly. Briefly explain the consequences if a diesel fuel is used in the S.I. Engine. 7

(c) What is ignition lag ? Discuss the effect of engine variables on ignition lag briefly. 7

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(d) An internal combustion engine is supplied with a mixture of octane vapour (C_8H_{18}) and air. Under steady running conditions, the dry exhaust gas analysis shows 14% carbon dioxide. Assuming complete combustion, determine the ratio by volume of fuel to air supplied, and express this as a percentage of the chemically correct ratio. Oxygen by volume is 21 percent in air. 7

Q. 3. (a) Discuss the drawbacks of a simple carburettor. 2

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(b) What do you mean by transient mixture requirement ? Discuss the mixture requirement for starting and warm up of the engine. Also discuss the function of choke system with suitable sketch. 7

(c) What is petrol injection system ? What are its merits and demerits ? 7

(d) The venturi of a simple carburettor has a throat diameter of 35 mm and the coefficient of air flow is 0.85. The fuel orifice has a diameter of 2.3 mm and the coefficient of fuel flow is 0.66. The petrol surface is 5 mm below the throat. Find :

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(i) The air fuel ratio for a pressure drop of 0.07 bar when nozzle lip is neglected.

(ii) The air fuel ratio for the pressure drop of 0.07 bar when nozzle lip is considered into account.

Take density of air and fuel as 1.2 kg/m^3 and 750 kg/m^3 respectively. 7

Q. 4. (a) What is firing order ? Name the factors that decides firing order of an engine. 2

(b) Describe two types of general injection system. Why the air injection system is not used nowadays ? Draw neat sketch of air injection system. 7

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(c) Discuss the functions of the Lubricating System. Write short note on Viscosity Index. 7

(d) Discuss the disadvantages of overcooling. Describe with a neat sketch thermostatically controlled forced circulation system. 7

Q. 5. (a) Define volumetric efficiency and relative efficiency. 2

(b) Discuss the various methods of control for exhaust emission from petrol engines. 7

(c) Describe the "Morse test". Make comments on 'assumptions made' & 'precaution taken' in performing the test. What is the accuracy of this test. 7

(d) In a trial of a single-cylinder oil engine working on dual cycle, the following observations were made :

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Compression ratio	=	15
Oil consumption	=	10.2 kg/hr
C.V. of fuel	=	43890 kJ/kg
Air Consumption	=	3.8 kg/min
Speed	=	1900 r.p.m.
Torque on brake drum	=	186 N-m
Quantity of cooling water used	=	15.5 kg/min
Temperature rise of cooling water	=	36°C
Exhaust gas temperature	=	410°C
Room temperature	=	20°C
C_p for exhaust gases	=	1.17 kJ/kg K

Calculate :

(i) Brake power

(ii) Brake specific fuel consumption

(iii) Brake thermal efficiency

Also draw heat balance sheet on minute basis. 7

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